

# Track D: Frontend Viewer, Validation & Demo

**Team:** 1 human + 1 AI agent **Owns:** `/track_d_frontend` **Starts Day 1** against Track C’s mock `/jobs/{id}` response and mock scaled mesh in `/shared/sample_jobs/job_0001` — do not wait for a real backend.

## 1. What this track delivers

- A simple web viewer: upload a photo, poll job status, render the returned 3D mesh (Three.js or equivalent), display dimensions.
- `accuracy_log.json` : for each hero image, the measured dimension from the pipeline vs. the human-supplied ground-truth measurement, and percent error, per the contract schema.
- A short accuracy report + a recorded fallback demo video, in case the live demo fails on presentation day.

## 2. Who does what, and why

**What the agent does:** - Writes the upload UI, polling logic, and 3D mesh rendering code. - Writes the accuracy-log computation and report generation. - Explains what a given rendering glitch likely means (e.g. wrong axis convention, wrong units, inverted normals) so the human’s bug report turns into a fast fix. - Drafts the demo script and narration, which the human then delivers.

**What the human does:** - Physically measures the real hero-image objects with a tape measure or equivalent, and supplies those ground-truth numbers — this is the entire basis of the accuracy report and can only come from a human in the room with the object. - Runs the viewer in an actual browser, on an actual device, and reports back what’s genuinely broken (mesh doesn’t load, orientation is wrong, units display incorrectly) versus what’s a minor polish item — a prioritization judgment call under time pressure. - Records the fallback demo video themselves (presenting is a human task), using the agent’s drafted script as a starting point, not a script to read verbatim. - Makes the final call on which hero images go into the demo — favoring the most visually convincing and most accurate results over including every test case, which is a presentation judgment, not a technical one.

**Why this split:** ground-truth measurement and live demo delivery are physical, human-only tasks; browser rendering can only be verified by actually opening a browser, which only the human can do and report back faithfully. The agent’s contribution is writing the viewer and report code quickly and diagnosing rendering bugs from the human’s description.

## 3. Day-by-day

| Day | Task                                                                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Agent scaffolds upload UI + polling + mesh renderer against the mock job response. Human confirms it renders the mock mesh at all (even if ugly). |
| 2   | Human measures real hero-image objects with a tape measure, logs ground truth; agent wires <code>accuracy_log.json</code> computation.            |
| 3   | Swap mock backend calls for Track C’s real (still-                                                                                                |

|   |                                                                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | synchronous) API as it comes online; human tests the full upload → render flow on real data end to end.                                                                      |
| 4 | Run all hero images through the live app; human records actual measured-vs-ground-truth numbers; agent generates the accuracy report; human records the fallback demo video. |
| 5 | Integration day — this track is the “front door” everyone will judge the project by; prioritize making 3 hero images look genuinely good over covering every edge case.      |

## 4. Definition of done

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- Live viewer works end-to-end on at least 3 real hero images through the real API.
- `accuracy_log.json` populated with real human-measured ground truth, not placeholder numbers.
- A recorded fallback demo exists, independent of the live app working on presentation day.

## 5. A note on the accuracy numbers

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Report the real percent error honestly, even if it’s large — per the earlier analysis, this MVP’s shortcuts (fixed focal length, aggressive downsampling, minimal mesh cleanup) are expected to cost real accuracy compared to the full 4-week plan. A demo that’s honest about “here’s our error rate and here’s exactly what we’d improve next” is stronger than one that hides it.